



THERMOCHEMISTRY

Chapter 11

Teaching Periods

8

Assessment

1

Weightage

9



Students will be able to:

- **Define** thermodynamics.
- **Define** the terms system, surrounding, boundary, state of system, state function, internal energy, enthalpy, entropy, heat of formation, standard heat of formation.
- **Classify** reactions as exothermic and endothermic.
- **Relate** change in enthalpy to the heat of reaction and heat of combustion of a reaction.
- **Relate** change in internal energy of a system with thermal energy at constant volume and constant pressure.
- **Explain** Hess's law with examples.
- **Apply** Hess's law to construct simple energy cycles.
- **Explain** reaction pathway diagram in terms of enthalpy changes of the reaction.

INTRODUCTION

Energy changes are the most important feature of a chemical process. When gasoline is ignited with air in a car engine, a certain amount of energy is released which some works in moving the piston due to which the wheels of the car rotate. Similarly, food can provide us with energy as it is burned up with oxygen in every cell of our body.

Heat and temperature are two different terms. Temperature describes the random movement of particles of a substance where as heat is associated with transfer of energy between two bodies with a temperature change. Chemical reactions are generally carried out with the transfer of heat to or from environment. **"The study of energy transference as heat during a chemical process is known as thermochemistry".**

11.1 THERMODYNAMICS

Thermodynamics is a Greek word, its first part 'Thermo' represents heat and the second part 'dynamic' which epitomizes flow or movement. **"Thermodynamics is an essential branch of science which deals with the study of inter transformation of heat, work, and other forms of energies"**. Thermodynamics applies to a variety of science such as physics, chemistry, engineering, etc and plays an imperative role in our daily life.

Thermodynamics is a macroscopic science and concerns with the initial and final states of a physical or chemical process but does not tell anything about the molecular constitution of matter. Hence, it helps to predict whether the thermochemical process can occur at particular conditions. However, it is incapable of assessing the rate of reaction.

11.1.1 Thermodynamics Terms

System:

A thermodynamic system is a quantity of matter of fixed identity, around which we can draw a boundary. **"It is a distinct part of our universe which is under particular experimental debate and separated from surrounding by an imaginary or real surface and can change its shape or size is called a system"**. It may be some water contained in a



beaker, some gas enclosed in a cylinder, or some solution filled in a calorimeter or the whole of the universe. All systems contain a certain quantity of matter and are characterized by particular macroscopic properties.

Surroundings:

Everything outside the arbitrarily selected boundaries of the system that affects the system or is affected by it is called surrounding. It is referred to as the remaining part of the universe other than a system from which energy or matter can be exchanged with the system. The system and surrounding together constitute the thermodynamic universe of that specific process.

Boundary:

It is a physical (real) or conceptual (imaginary) closed surface that separates a system from its surroundings.

For example if a reaction between zinc dust and dilute hydrochloric acid is carried out in a container.

- The contents of the container, (zinc and HCl) constitute the system because they are characterized by specific factors, like mass, volume, temperature, etc.
- The entire remaining part of the universe including the wall and empty part of the container specifies its surrounding.
- The surface of the reacting material which separates the system from its surrounding is said to be a physical boundary.

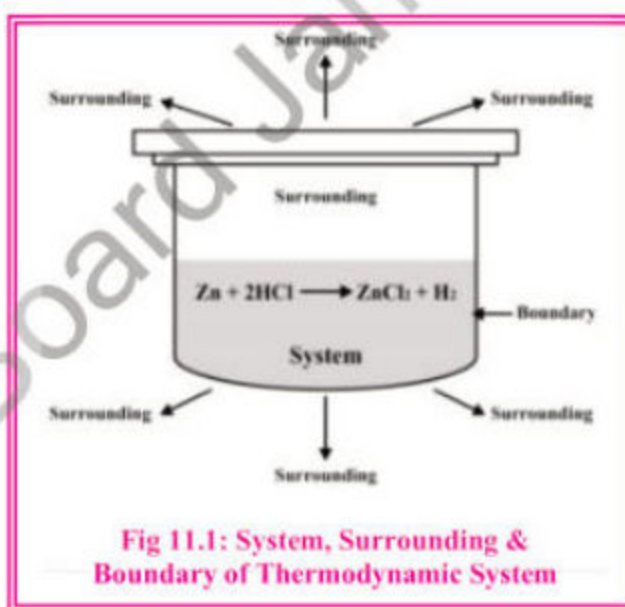


Fig 11.1: System, Surrounding & Boundary of Thermodynamic System

Thermodynamic systems are categorized into three main types on the basis of their ability to allow an interchange of mass and energy into environment through their boundaries.

Open System:

All such thermodynamic systems which communicate with the surrounding in terms of both mass transfer and energy transfer are known as open systems. The boundary of this type of system is open and non-insulated.

A boiling tea kettle is an open system because it transfers heat and matter (steam) to the surrounding.

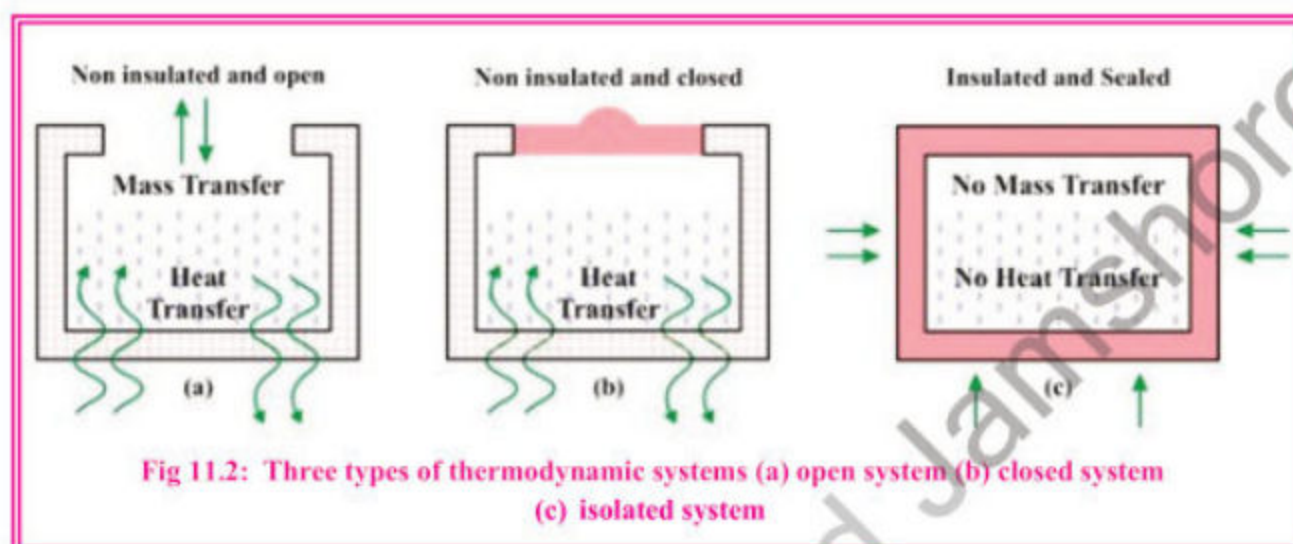
Closed System:

All such thermodynamic systems which do not allow the exchange of mass but permit energy to be transferred with the environment are known as closed systems. The boundary of a closed system is sealed but not insulated hence it is impermeable to mass only.

A pressure cooker is a good approximation to a closed system where only heat transfers (except a little steam escapes through the top valve to prevent explosion).



Think of any other close systems around you!



Isolated System:

All such thermodynamic systems which neither exchange matter nor energy with the surrounding and hence totally independent of their surrounding are known as an isolated systems. The boundary of an isolated system is sealed and insulated.

A perfect isolated system does not exist as energy interaction in the form of heat radiation will always occur. However, some conceptually true isolated systems are thermo flask and insulated drink cooler.

Macroscopic Properties:

The description of a thermodynamic system is associated with some macroscopic quantities. These quantities can be measured if the system has large number of particles, thus macroscopic properties refers to the system having bulk amount. These are further classified into extensive and intensive properties depending upon whether they are proportional to the size of the system or not.

Entropy (S):

It is a measure of the disorder of the system. For example the arrangement of particles are more disordered in liquid and gases than solids. It is directly proportional to temperature.

State of System:

A thermodynamic system is said to be in a certain state if some macroscopic properties such as temperature pressure, volume, and moles are known. With the change of one or more of these macroscopic properties, the system changes its state.



Do You Know?

Intensive properties are those whose values are not dependent on quantity of matter in the given system. For example

Pressure, density, Refractive index, surface tension, Melting point etc.

Extensive Properties are those whose values for the entire system depend on the quantity of matter. For example

Mass, Volume, Energy, Enthalpy, Moles etc.



The description of a system before the change occurs is known as an initial state whereas it is said to be a final state when the change has occurred. The change in any macroscopic properties of the system is determined by taking the difference of their values at the final and initial state.

For example in a chemical reaction, the enthalpy of reactant is represented by H_1 (initial state), and the enthalpy of product is shown by H_2 (final state) hence $(\Delta H^\circ = H_2 - H_1)$ is considered as the change in enthalpy in this chemical process.

State Function:

A state function is a thermodynamic parameter which is dependent on initial and final state of the system and is independent of how the change is occurred. Examples of state function are internal energy, enthalpy, temperature, volume etc.

Enthalpy of a System:

Heat transfer in almost all physical and chemical processes happens at constant volume. In the laboratory, for example, the reaction is usually carried out in open test tube, beakers or flask etc and hence the heat flow into or out of the reaction generally takes place under the condition of the constant pressure of air (approximately 1 atmosphere). In this situation heat associated with the system may utilize both in altering the internal energy as well as performing work which may be seen by changing the volume of the reaction mixture in the reaction container.

To quantify the total heat contents of a system, chemists use a quantity named enthalpy (H). It defines as **“The sum of internal energy and algebraic product of its pressure and volume”**.

$$H = E + PV \text{ ----- (i)}$$

Equation (i) specifies that enthalpy of a system depends on the quantity of internal energy, pressure and volume. Since P , V and E are all state functions, we must say that enthalpy is also a state function.

Enthalpy change (ΔH) of a reaction is measured in Calorie (metric unit system) as well as Joule (S. I unit system). The interrelation between Joule and calorie is given as

$$1 \text{ Cal} = 4.18 \text{ Joules}$$

Experience shows that in many chemical reactions amounts of heat evolve or absorb in thousands of joules per mol. Hence for convenience it is measured in KJ/mol.



Self Assessment

- What is meant by enthalpy of system? Mention its units.
- Which of the following are not classified into state function.
Work (W), Enthalpy change (ΔH), Volume change (ΔV), Internal energy change (ΔE).



11.2 THERMOCHEMICAL REACTIONS

The progress of a chemical reaction i.e. the conversion of reactants into products involves the breaking of existing bonds of reactant particles for which energy has to be supplied and the formation of new bonds in which energy is released. Indeed, different bonds have different energy requirements for their breaking and re-forming thus to decide whether a chemical reaction takes place with the absorption or evolution of heat, we must know the energy difference of reactants and products. Since chemical reactions are usually taking place at constant pressure and the enthalpy of a system is stored in the chemical bond, the energy change in the formation of products from reactants is represented by ΔH .



Do You Know?

Gas welding is a combustion reaction between acetylene and oxygen. When the two gases contact to each other in a hand-held torch a relatively hot flame is produced having a temperature of 3000°C .

11.2.1 Exothermic Reactions

Exothermic reactions (Exo; out) are those in which heat is given out making the surrounding warmer. In the course of exothermic reactions, energy stored in the chemical bond of products is less than reactant hence these reactions are energetically favorable and often occur spontaneously but sometimes we need extra energy to get them started. The overall energy released in these reactions is represented by ΔH with a negative sign.

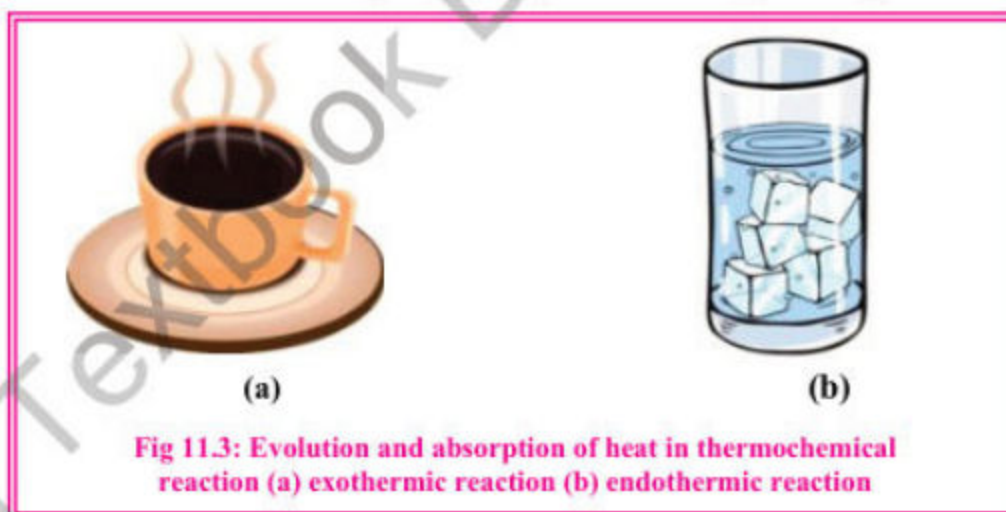
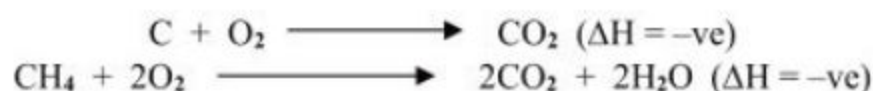


Fig 11.3: Evolution and absorption of heat in thermochemical reaction (a) exothermic reaction (b) endothermic reaction

All combustion reactions including the burning of fuel and coal, oxidation of Sui gas, etc are exothermic reactions and can easily be noted by a rise of the surrounding temperature.



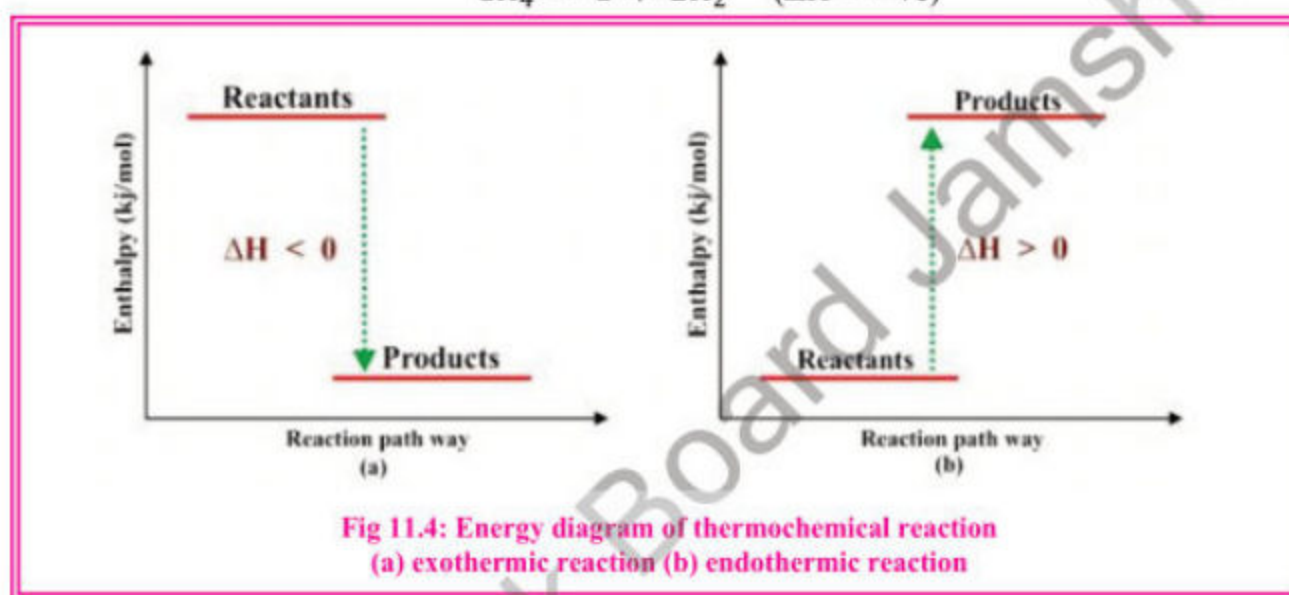
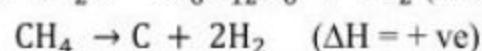
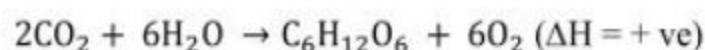
11.2.2 Endothermic Reactions

Endothermic reactions (endo; taking in) are those in which heat is put into the system making the surrounding colder. In these reactions more energy is needed to break the bonds in the reactants than is released when new bonds are formed in the products. Since heat is used



for the reaction to occur, these are referred to as non-spontaneous reactions. The net energy absorbed in these reactions is represented by ΔH with a positive sign.

An appropriate endothermic reaction is photosynthesis during which plants absorb solar heat through their chlorophyll parts and this heat is utilized for the conversion of CO_2 and H_2O into glucose.



Self Assessment

Classify each of the following processes as exothermic or endothermic.

- | | |
|--------------------------|-------------------------------------|
| (i) Synthesis of Ammonia | (ii) Decomposition of Lime |
| (iii) Oxidation of Coal | (iv) Neutralization of NaOH and HCl |
| (v) Combustion of Hexane | (vi) Burning of Sui gas |

11.3 FIRST LAW OF THERMODYNAMICS

The first law of thermodynamic is an adaptation of the law of conservation of energy that governs the transference of energy between the system and surrounding in the form of heat and work. This concept was enunciated by J.R. Mayer (1841) and formulated by Helmholtz (1847).

11.3.1 Statement and Mathematical Derivation

This law may be stated as; “In any physical or chemical process, the net energy of a system and its surroundings must remain constant”. Or “a system can not create or destroy energy, however, it can exchange energy with its surrounding in the form of heat and work”. This law describes us that heat and work are mutually convertible but since energy can neither be created nor be destroyed, the total energy conversion remains constant.



To develop the mathematical approach of the law let us consider a system in its initial state having internal energy E_1 . If “ q ” be the heat supplied to the system while W be the work done on the system, the internal energy increases to E_2 . Then according to the law, the internal energy change ($E_2 - E_1$) may be formulated as

$$\Delta E = q + w \quad \text{----- (iii)}$$

Where

q = heat transfer across the system

W = work done

ΔE = change in the total energy of the system ($E_2 - E_1$)

The sign of ‘ q ’ is taken positive if thermochemical process is concerned with heat absorption but in case of heat release it is taken as negative sign. The sign of W should be taken negative for thermochemical expansion and positive if it undergoes compression.

Equation (iii) may take the form $\Delta E = q$ if the heat is transferred to the system without work is being done but if work is done on the system without heat transfer it may change to $\Delta E = W$.

11.3.2 Derivation of Pressure-Volume Work Equation

Mechanical work due to expanding or squeezing of a system at constant external pressure is known as pressure-volume work.

Consider a gas confined in a cylinder fitted by a negligible weight, frictionless and movable piston of cross-section area A at a constant external pressure. If the force exerted by the gas on the inner wall of the piston is greater than external pressure, the piston moves upward from a height h_1 to h_2 and hence does some work which may be formulated as

$$W = -F(h_2 - h_1)$$

Insertion of a negative sign indicates that external pressure opposes the expansion of gas. Since pressure is the force exerted on unit area ($P = F/A$) and thus the force is ($F = PA$), the work may be written as

$$W = -P A (\Delta h)$$

But $A \cdot \Delta h$ represents volume change (ΔV) of the system so work done on thermodynamic system is deduced as

$$W = -P \Delta V \quad \text{----- (iv)}$$

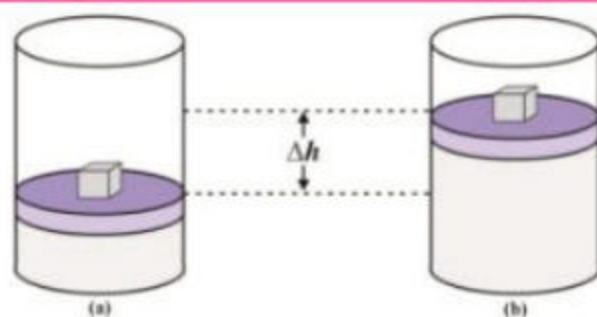


Fig 11.5: pressure-volume work in thermodynamic expansion process (a) volume at initial state (b) volume at the final state



As far as the unit of pressure-volume work is concerned, we may have two options; either it takes in Joules (J) or in $\text{atm} \cdot \text{dm}^3$. In S.I measurement pressure is taken in Pascal (N/m^2) and volume in cubic meter (m^3), their product will be in Joules. However, it is more agreeable to express pressure in atmosphere and volume in dm^3 , it makes the unit of work is $\text{atm} \cdot \text{dm}^3$. The two work units are co-related as

$$1 \text{ atm} \cdot \text{dm}^3 = 101.325 \text{ Joules}$$

Example 11.1

A gaseous chemical reaction is carried out in a cylinder under a constant external pressure of 1 atm. If during the reaction volume increases from 3 dm^3 to 5 dm^3 by moving the piston upward. Calculate the workdone and express in kJ.

Solution:

Since the work done in an expansion process has negative sign, the equation of pressure volume work should be $W = -P\Delta V$.

$$\text{Work} = -P\Delta V$$

$$\text{Work} = -1 \text{ atm} (5 \text{ dm}^3 - 3 \text{ dm}^3)$$

$$\text{Work} = -2 \text{ atm dm}^3$$

But $1 \text{ atm dm}^3 = 101.325 \text{ J}$, hence

$$\text{Work} = -202.65 \text{ J} = -0.20265 \text{ KJ}$$



Self Assessment

Convert the values of following quantities.

- 20 calories energy into Joule
- 3.5 atm dm^3 work into KJ.

Internal Energy

Every system is made of particles (atoms, ions, or molecules) and has a specific amount of energy. This energy is associated with the sum of all microscopic energies of the particles of the system and mainly categorised as kinetic and potential energies. The kinetic energy comprises of particles motion such as translational, vibrational, rotational motions etc. potential energy includes attractive and repulsive interaction between particles and other chemical energies stored in the chemical bond. **"The total energy content of a thermodynamic system under a given set of conditions is known as internal energy".**

Change in internal energy (ΔE) can be measured experimentally ($\Delta E = E_2 - E_1$). Internal energy change (ΔE) is a state function because it depends only upon the initial and final state and not by a path to which change occurs.



Example 11.2

Burning of petrol in an automobile engine gives carbondioxide and water vapours. If the gases do 675J work in pushing the piston outward and the system lose 435J heat to the surrounding, calculate the internal energy change in kJ.

Solution:

Since heat is given out, the value of q is taken with negative sign. Likewise sign of W is also taken negative because work is done by the system.

$$q = -435J$$

$$W = -675J$$

$$\Delta E = ?$$

$$\Delta E = q + w$$

$$\Delta E = (-435) + (-675)$$

$$\Delta E = -1110J$$

$$\Delta E = -1.11KJ$$

11.3.3 Applications at Constant Volume and Constant Pressure

So far we have learned categorically that heat absorb or evolve in a chemical process is related to its internal energy and PV work done. Now we will see that how the equation of First law of thermodynamics is applicable to the process that occur under constant pressure and constant volume.

The process at Constant Volume

When a reaction is carried out at constant volume ($\Delta V = 0$) no work is done on or by the system. In this situation internal energy change must equal to heat change of the reaction. The equation of the First law of thermodynamics may be reformed as $q_v = \Delta E$ (the subscript "v" specify constant volume process).

$$q_v = \Delta E \text{ ----- (v)}$$

The process at Constant Pressure

It is a matter of fact that many chemical reactions in labs or workshops are carried out in open vessels. This means they carried out at constant pressure in a manner of thermochemical expansion. Thus according to the first law of thermodynamics, this constant pressure process may be illustrated as

$$\Delta E = q_p - P\Delta V$$

$$\text{or } q_p = \Delta E + P\Delta V \text{ ----- (vi)}$$

Where q_p is the heat flow to or from the system at constant pressure, ΔE is internal energy change, and $P\Delta V$ is work done with changing the volume.

$$q_p = (E_2 - E_1) + P(V_2 - V_1)$$

$$q_p = (E_2 + PV_2) - (E_1 + PV_1)$$

$$q_p = H_2 - H_1$$



And $q_p = \Delta H$ ----- (vii)

Equation (vii) tells that, if a chemical process is occurred at constant pressure where only PV work is allowed due to compression or expansion the enthalpy change (ΔH) of system is equal to energy flow (q).

Substituting this value in equation (vi) we get

$$\Delta H = \Delta E + P\Delta V \text{ ----- (viii)}$$

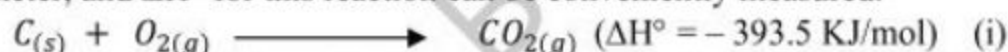
Equation (viii) represents gaseous reaction. However, for the reaction in which one or more solids or liquids are involved, the value of $P\Delta V$ becomes almost zero because solids and liquids undergo very small volume change. Thus in this case enthalpy change (ΔH) will be equal to internal energy change (ΔE) and the equation of First law of thermodynamics is reformed as

$$\Delta H = \Delta E \text{ ----- (ix)}$$

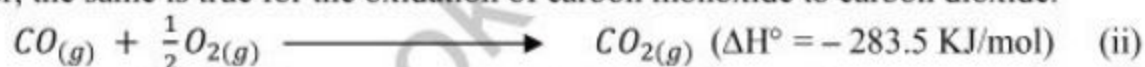
11.4 HESS'S LAW: (Enthalpy change calculations)

The change in enthalpy of some reactions are measured by calorimetric method. However, experimentally it is difficult to determine change in enthalpy of very slow and very fast reaction.

For example, combustion of carbon (graphite) to give carbon dioxide can be worked in a calorimeter, and ΔH° for this reaction can be conveniently measured.



Further, the same is true for the oxidation of carbon monoxide to carbon dioxide.

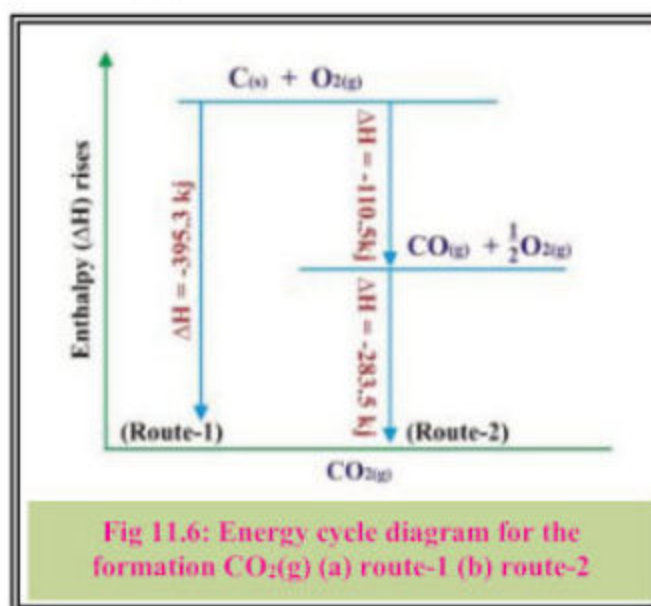


Contradictorily, the measurement of ΔH° in the combustion of carbon to carbon monoxide is quite difficult because some of the carbon monoxides is further oxidized to carbon dioxide. (Subtracting equation ii from i)



Enthalpy change of a reaction is a state function and it depends only on where the chemical process begins and finishes, but not on the route it takes from. Hence considering the energy cycle of the above three reactions (illustrated in fig 11.6), ΔH° of carbon monoxide can be measured indirectly.

G.H. Hess (1840) introduced a law of thermochemistry based on his observations on cyclic thermochemical changes which involve two or more different routes. This is known as Hess's law of enthalpy summation.

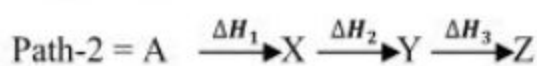




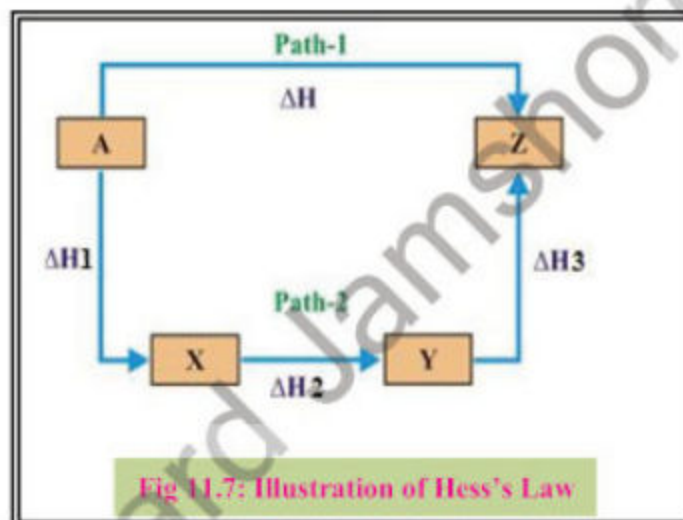
This law states as “if a chemical reaction can be brought about in more than one path way, the net enthalpy change is the same provided that the initial and final states are the same”.

11.4.1 Explanation of Hess's Law

Let us suppose a substance A is converted to Z by two different paths (illustrated in fig 11.7). It may either change directly to Z in one step or by involving some intermediate steps in which X and Y are formed. In both cases the total enthalpy change remains the same.



$$\text{Thus } \Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$$



Standard Enthalpies of Formation

It defines as “the enthalpy change of a chemical reaction in which one mole of a pure substance is formed from its elements with all substance in their standard state”.

The standard enthalpy of formation is represented by the symbol ΔH_f° in which superscript zero signifies standard conditions of 1 atmospheric pressure and 25°C, whereas subscript “f” stands for formation.

Conventionally ΔH_f° of most of the elements (H_2 , O_2 , N_2 , Na, etc) are taken as zero because no further change is needed to bring them to standard state condition. However, for those elements which exists in more than one allotropic forms, the most stable allotropic form is conventionally assumed to have $\Delta H_f = 0$. For example, graphite is a thermodynamically more stable allotropic form of carbon than diamond at 25°C and 1 atm pressure hence ΔH_f of graphite is taken as zero. Similarly, O_2 (oxygen) is a more stable allotropic form than O_3 (ozone), and therefore ΔH_f of O_2 is assumed to be zero. The same case in for Rhombic sulfur ($\Delta H_f = 0$)

Table 11.1 The Standard States of various substances	
Substances	Standard States
All pure solids & liquids	25°C
All gaseous compounds	25°C and 1 atm
All substances in the solution	25°C and 1 molar conc.
All elements	25°C and 1 atm



Standard Enthalpy of Reaction

It is the enthalpy absorbed or evolved when fixed number of moles of reactants as represented in a balance chemical equation are completely reacted to form product with the condition that all reactants and products are in their standard states. It is represented by ΔH° . In general ΔH° in a balanced thermochemical equation is the sum of ΔH_f° of products minus the sum of ΔH_f° of reactants.

$$\Delta H_{\text{reaction}}^\circ = \left[(\Sigma np \Delta H_f^\circ(\text{products})) - (\Sigma nr \Delta H_f^\circ(\text{reactants})) \right]$$

Where np and nr represent total number of moles of reactants and products take part in the reaction.

Table 11.2 Standard Molar Enthalpy of Formation of some organic and inorganic substances at 25°C (In KJ/mol)			
Organic substances	ΔH_f° (KJ/mol)	Inorganic substances	ΔH_f° (KJ/mol)
CH _{4(g)}	– 74.85	CO _(g)	– 110.5
C ₂ H _{6(g)}	– 83.8	CO _{2(g)}	– 393.5
C ₃ H _{8(g)}	– 103.9	SO _{2(g)}	– 296.8
C ₂ H _{4(g)}	+ 52.30	SO _{3(g)}	– 396
C ₂ H _{2(g)}	+ 226.7	H ₂ S _(g)	– 20.17
C ₆ H _{6(l)}	+ 49.03	NH _{3(g)}	– 46.19
CH ₃ OH _(l)	– 239	H ₂ O _(l)	– 285.8
C ₂ H ₅ OH _(l)	– 277.7	Fe ₂ O _{3(s)}	– 824.2
CH ₃ COOH _(l)	– 484.2	NaCl _(s)	– 411

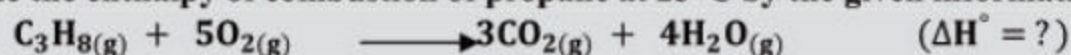
Use of Hess's Law in the determination of heat of reaction

There are two ways to calculate ΔH° of a reaction. It either determined calorimetrically or calculate from known values of ΔH_f° of all the substances involve in the reaction. Hess's law provides us a convenient way to find out the heat change involves in a chemical reaction by subtracting the sum of heat of formation of all reactants from the sum of heat of formation of all products.



Example 11.3

Calculate the enthalpy of combustion of propane at 25°C by the given informations.



$$\Delta H_f^\circ \text{ of } \text{C}_3\text{H}_{8(g)} = -103.9 \text{ KJ/mol}$$

$$\Delta H_f^\circ \text{ of } \text{CO}_{2(g)} = -393.5 \text{ KJ/mol}$$

$$\Delta H_f^\circ \text{ of } \text{H}_2\text{O}_{(l)} = -285.8 \text{ KJ/mol}$$

Solution:

The mathematical form of heat of reaction ΔH° is given as.

$$\Delta H_{\text{reaction}}^\circ = \left[(\sum n_p \Delta H_f^\circ(\text{products})) - (\sum n_r \Delta H_f^\circ(\text{reactants})) \right]$$

Substituting the values.

$$\Delta H_{\text{reaction}}^\circ = \left[(-393.5 \times 3) + (-285.8 \times 4) \right] - \left[(-103.9 \times 1) \right]$$

$$\Delta H_{\text{reaction}}^\circ = -2219.8 \text{ KJ/mol}$$

Use of Hess Law in the determination of Heat of formation

Realizing the fact that thermochemical equations can be added subtracted or multiplied like ordinary algebraic equations, Hess's law is useful to calculate the heat of formation of many compounds where an experimental determination is not possible.

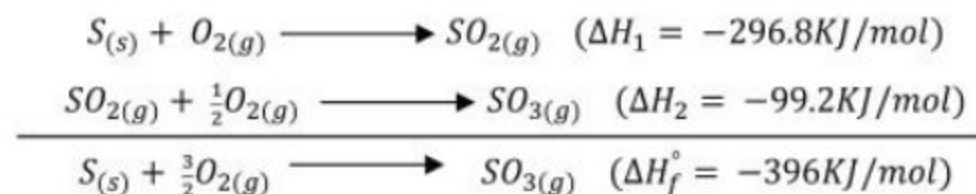
To understand how is Hess's Law useful in thermochemistry, look at the oxidation of sulphur to sulphur trioxide.



Analyzing the reaction, we find that sulphur can not be directly oxidized into SO_3 , therefore ΔH_f° of SO_3 can not be determined if we put sulphur and oxygen in a calorimeter. The only option is to go for an alternative route which consists of following two steps.



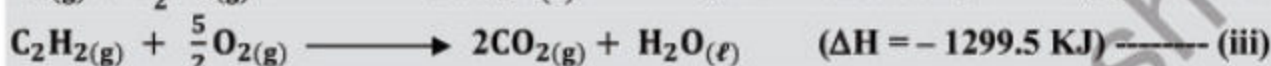
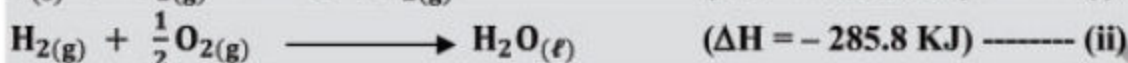
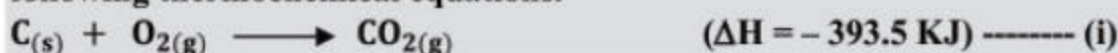
Thus the ΔH_f° of $\text{SO}_{3(g)}$ is indirectly determined by adding enthalpy change of step 1 and step 2.



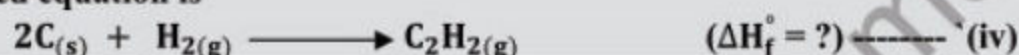


Example 11.4

Calculate the standard heat of formation of acetylene (C_2H_2) by using the data of the following thermochemical equations.



The desired equation is



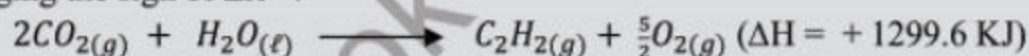
Solution:

To determine ΔH_f° value of C_2H_2 , we have to generate the desired equation by manipulating and adding eq. (i), (ii) and (iii) together in such a way that all unwanted substances should be canceled.

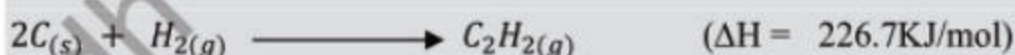
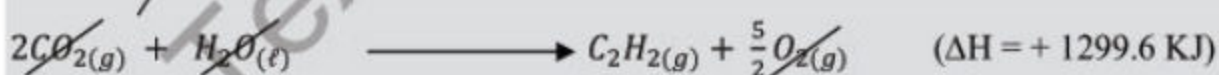
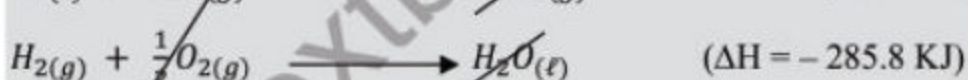
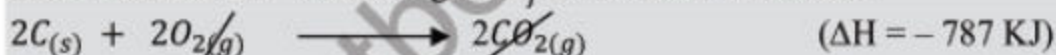
In the desired equation (iv) we see that 2 moles of carbon and 1 mole of H_2 are needed as a reactant, this can be obtained by multiplying equation (i) by 2, which means that ΔH° will also be multiplied by 2.



Next, we see that C_2H_2 is needed as a product. To do this, the equation (iii) must be reverse with changing the sign of ΔH° .



Now sum up these three manipulated equations to get the required reaction. Also, sum up the three modified ΔH° values to get ΔH_f° of the desired reaction.



11.4.2 Born Haber Cycle

Born Haber cycle is developed by two German chemists Max Born and Fritz Haber (1919) to correlate the lattice energy of ionic solids with some other energies involved in their formation. **"It is a cycle of enthalpy change in which an ionic solid is theoretically formed from its elements in their standard states"**. It is a



Max Born



Fritz Haber



specific application of Hess's law and generally use to determine lattice energy of ionic solids.

To understand, how the lattice energy of a binary ionic solid determines by using Born Haber cycle let's take solid Cesium fluoride (CsF) as an example. We start by writing a thermochemical equation for the formation of CsF from its elements at their standard states (25°C and 1 atm).

The standard enthalpy of formation of $\text{CsF}_{(s)}$ as mentioned in table 11.2 is -553.5 KJ/mol

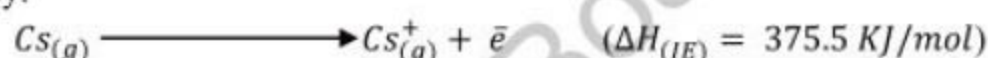


Now various energies involved in the formation of $\text{CsF}_{(s)}$ are described in the following stepwise manner.

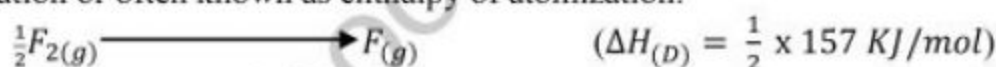
Step-1: Sublimation of Cesium It involves the vapourization of metallic cesium into gaseous atom. The process is endothermic because heat is always needed to sublime a solid.



Step-2: Ionization of Cesium atom It involves the loss of electron from cesium gaseous atom. The enthalpy change corresponds to this process is always positive and known as ionization enthalpy.



Step-3: Dissociation of gaseous Fluorine It involves the breaking of F_2 molecule into fluorine gas atoms. The enthalpy related to this process is ever positive and labeled as enthalpy of dissociation or often known as enthalpy of atomization.



Step-4: Formation of $\bar{\text{F}}$ ion It involves the absorbition of electron into fluorine gas atom. The enthalpy change of this thermochemical process is negative and specify as electron affinity.



Step-5: Formation of Solid CsF It involves the electrostatic interaction of oppositely charged gaseous ions. The enthalpy change in this process is indeed negative and known as Lattice energy.



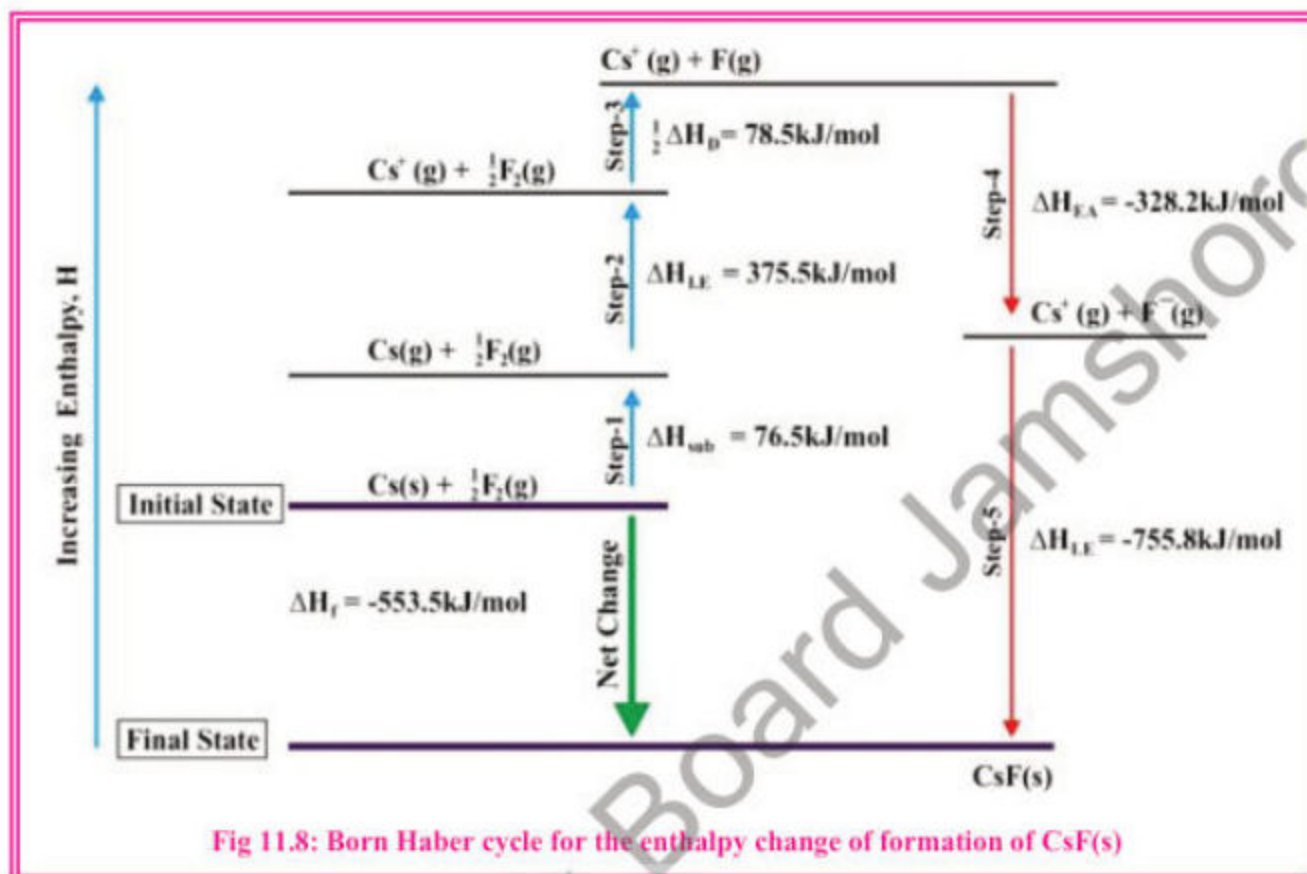
Intended with Hess's Law, the sum of enthalpy change of these five steps will be equal to the enthalpy of formation of $\text{CsF}_{(s)}$ being the overall enthalpy change of the energy cycle is zero. This may be set out in fig 11.10 and formulated as.

$$\Delta H_f^\circ = \Delta H_{sub} + \Delta H_{IE} + \frac{1}{2}\Delta H_D + \Delta H_{EA} + \Delta H_{LE}$$

Substituting the values of concerned energies, we get lattice energy.

$$(-553.5) = (76.5) + (375.5) + \left(\frac{1}{2} \times 157\right) + (-328.2) + (-\Delta H_{LE})$$

$$\Delta H_{LE} = -755.8\text{ KJ/mol}$$



Example 11.5

Use the data provided below for the formation of $\text{RbCl}_{(\text{s})}$, write thermochemical equations for all the steps involved in the Born Haber cycle and determine the enthalpy of formation of $\text{RbCl}_{(\text{s})}$.

Sublimation energy of $\text{Rb}_{(\text{s})} = 82 \text{ KJ/mol}$

Ionization energy of $\text{Rb}_{(\text{g})} = 403 \text{ KJ/mol}$

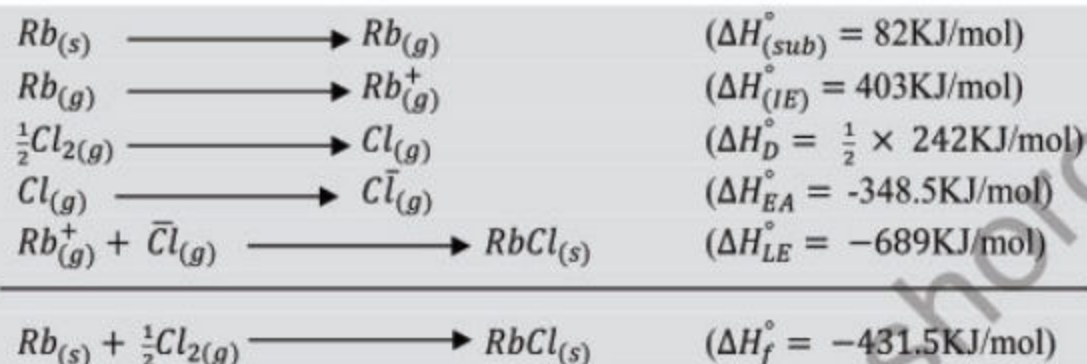
Dissociation energy of $\text{Cl}_{2(\text{g})} = 242 \text{ KJ/mol}$

Electron affinity of $\text{Cl}_{(\text{g})} = -348.5 \text{ KJ/mol}$

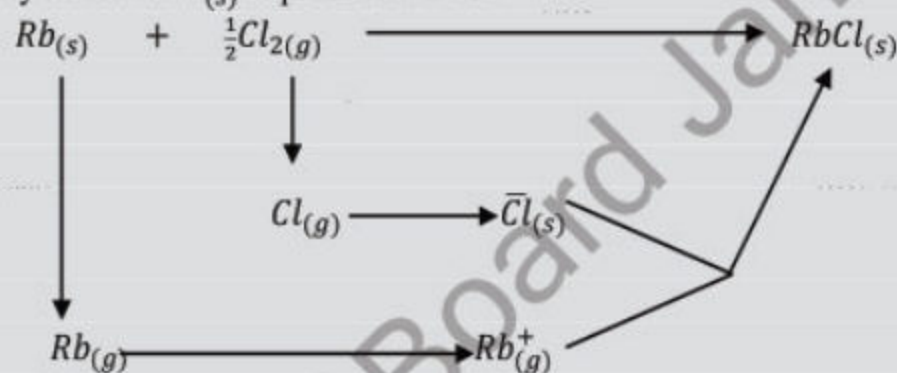
Lattice energy of $\text{RbCl} = -689 \text{ KJ/mol}$

Solution:

Since, according to Hess's law overall enthalpy change of a cyclic thermochemical process is zero, the sum up of all five steps involved in the process will get heat of formation of RbCl . Thermochemical equations associated with Born Haber cycle in the formation of $\text{RbCl}_{(\text{s})}$ maybe written as



The Born Haber cycle on $\text{RbCl}_{(s)}$ is presented below



Heat of formation of $\text{RbCl}_{(s)}$ may also be calculated by summing up the individual steps involved in the following simple way.

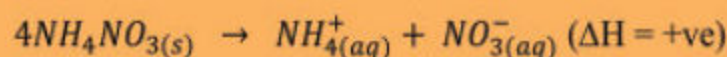
$$\begin{aligned}
 \Delta H_f^\circ &= \Delta H_{sub} + \Delta H_{IE} + \frac{1}{2} \Delta H_D + \Delta H_{EA} + \Delta H_{LE} \\
 \Delta H_{f(\text{RbCl})}^\circ &= (82) + (403) + \left(\frac{1}{2} \times 242\right) + (-348.5) + (-689) \\
 \Delta H_{f(\text{RbCl})}^\circ &= -431.5 \text{ KJ/mol}
 \end{aligned}$$



Society, Technology and Science

Use of cold pouche for cooling

Endothermic reactions are all around us. You might have sprained on your wrist or swollen on the elbow, in these circumstances you often put an instant cold pack for quick relief. Let's imagine, what happens. The bag consists of ammonium nitrate and water in separate packing and when torn, they get mixed. The polar terminals of water molecules interact with ionic bond of ammonium nitrate and the energy needed for this process is absorbed from the surrounding which makes the water colder and thus bag feels chill.





Activity

You will always find a kitchen having usual cooking items such as vinegar (CH_3COOH) and baking soda (NaHCO_3). Can you deduce whether the chemical reaction between them is exothermic or it is an endothermic one?

For that, you need to have a glass tumbler and pour very little amount of acetic acid in it ($1/5$ of the volume). Check the temperature initially through a temperature probe (the one which you use to note the temperature of coffee). Let it be dipped in the tumbler. Now add a teaspoon of baking soda in it and keep yourself at a distance. You will observe bubbles forming rapidly. It appears as if something is boiling and your mind could misconceive it as an exothermic reaction but the temperature probe would show you a drop in temperature indicating that in actual it is an endothermic reaction.

SUMMARY with Key Terms

- ◆ **Thermodynamics** is the branch of physical science which deals with the study of intertransformation of heat work and other forms of energies.
- ◆ **Thermochemistry** is the study of energy change associated with a physical or chemical process.
- ◆ **Thermodynamic System** is a distinct part of the universe that is under experimental debate and separated from surrounding by a boundary.
- ◆ **Internal Energy, E** is mainly associated with the sum of all kinetic and potential energies of the particles of a system under study.
- ◆ **Enthalpy, H** is the heat contents of a system under constant pressure and describe as $H = E + PV$.
- ◆ **State Function** is a thermodynamic parameter on which description of a system is made, it depends on the initial and final state of the change but not on the path.
- ◆ **Exothermic Reactions** are those in which heat is given out making the surrounding warmer.
- ◆ **Endothermic Reactions** are those in which heat is put into the system making the surrounding colder.
- ◆ **The First Law of Thermodynamics** states that the total energy of a system and its surroundings must remain constant.
- ◆ **Pressure-Volume Work** is mechanical work associated with expansion or compression of a thermodynamic system, under a constant external pressure.
- ◆ **Standard Enthalpy of Formation, ΔH_f°** is the heat evolve or absorb when 1 mole of a substance is produced from its constituent elements under their standard states i.e 25°C and 1 atm pressure.
- ◆ **Standard Enthalpy of combustion ΔH_c°** heat energy given out when one mole of a substance burn completely in air.



- ◆ **Standard Enthalpy of Reaction, ΔH°** is the heat change of a chemical reaction and can be measured by subtracting the sum up values of ΔH_f° of products to the sum up values of ΔH_f° of reactants.
- ◆ **Hess's Law** tells that if a reaction is carried out through several steps then the enthalpy change for the overall reaction must be equal to the sum of enthalpy change of these individual steps.
- ◆ **Born Haber Cycle** is an application of Hess's Law and generally use to determine the Lattice of ionic compounds.

EXERCISE

Multiple Choice Questions

1. Choose the correct answer

- Least entropy found in which of the following state of water:
(a) Liquid water at 4°C (b) Liquid water at 25°C
(c) Steam at 100°C (d) Ice at 0°C
- The energy corresponds to the given thermochemical process is labeled as:
$$\text{Li}_{(g)}^+ + \bar{\text{Cl}}_{(g)} \longrightarrow \text{LiCl}_{(s)}$$

(a) Ionization energy (b) Lattice energy
(c) Enthalpy of combustion (d) Enthalpy of atomization
- Which of the following change is not an endothermic reaction:
(a) Decomposition of lime (b) Cracking of alkanes
(c) Combustion of butane (d) Photosynthesis
- Heat transfer can not be feasible across the boundry of a:
(a) Open system (b) Close system
(c) Isolated system (d) Thermo permeable system
- Which of the following enthalpy change is always negative:
(a) Enthalpy of formation (b) Enthalpy of decomposition
(c) Enthalpy of combustion (d) Enthalpy of reaction
- Standard enthalpy of formation of all of the following elements at 25°C and 1 atm pressure are zero except:
(a) $\text{C}_{(\text{diamond})}$ (b) $\text{C}_{(\text{graphite})}$
(c) O_2 (d) N_2
- In the equation of First law of thermodynamics ($\Delta E = q + w$), the property(s) which depends upon initial and final state is (are):
(a) ΔE (b) q
(c) W (d) Both q and W



- (viii) Which statement is incorrect:
- (a) For constant pressure process, $\Delta H = \Delta E + P\Delta V$
 - (b) For constant volume process, $\Delta E = q$
 - (c) For exothermic reactions, $\Delta H > 0$
 - (d) For Hess law $\sum \Delta H^\circ_{\text{(cycle)}} = 0$
- (ix) In a thermochemical process, no work is done if the system is kept at:
- (a) Constant temperature
 - (b) Constant pressure
 - (c) Constant volume
 - (d) Constant mass
- (x) Volume is a:
- (a) State function
 - (b) Path function
 - (c) Intensive properties
 - (d) Colligative property

Short Questions

1. State precisely the meaning of each of the following terms.
 - (i) System and surrounding
 - (ii) State and state function
 - (iii) Internal energy and Enthalpy
2. What is meant by Internal energy change (ΔE) and Enthalpy change (ΔH)? Under what conditions are ΔE and ΔH equal?
3. How can you define standard enthalpy of formation and standard enthalpy of reaction.

Descriptive Questions

1. State and explain First law of Thermodynamics. Derive pressure-volume work of a system.
2. Discuss the applications of the First law of thermodynamics at constant pressure and constant volume.
3. State and explain Hess's Law of enthalpy summation. Discuss its applications.
4. Explain Exothermic and Endothermic reactions with the help of the energy diagram.

Numerical Questions

1. A thermochemical process is carried out at constant pressure of 8.52 atm. If it absorbs 15.4 KJ energy from the surrounding due to which an expansion in the volume of 4.7 dm³ is occurred. Calculate its change in internal energy. [Ans: 11.34 KJ]
2. Using the data in table 11.2, calculate the standard enthalpy change for each of the following reactions.





3. In the manufacturing of HNO_3 by the Ostwald process, one of the most important exothermic reactions is the oxidation of ammonia.



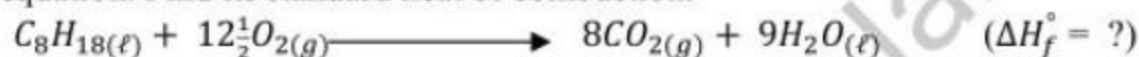
Determine standard heat of reaction (ΔH°) from the following given data.

$$\Delta H_f^\circ \text{ of } \text{NH}_{3(g)} = -46.19 \text{ KJ/mol}$$

$$\Delta H_f^\circ \text{ of } \text{NO}_{(g)} = 90.25 \text{ KJ/mol}$$

$$\Delta H_f^\circ \text{ of } \text{H}_2\text{O}_{(l)} = -285.8 \text{ KJ/mol} \quad [\text{Ans: } -1169.04 \text{ KJ/mol}]$$

4. Iso octane (C_8H_{18}) is an efficient fuel with a high octane rating. The combustion of C_8H_{18} in an internal combustion engine is represented in the following thermochemical equation. Find its standard heat of combustion.

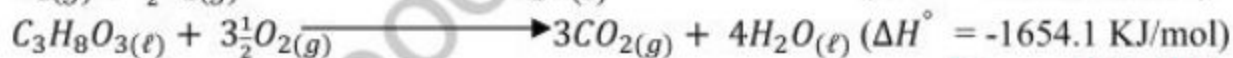
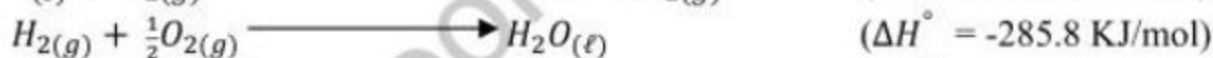
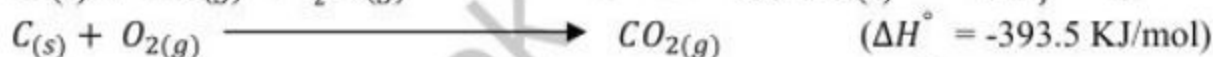


Given that $\Delta H_f^\circ \text{ of } \text{CO}_2 = -393.5 \text{ KJ/mol}$

$$\Delta H_f^\circ \text{ of } \text{H}_2\text{O} = -285.8 \text{ KJ/mol}$$

$$\Delta H_f^\circ \text{ of } \text{C}_8\text{H}_{18} = -223.8 \text{ KJ/mol} \quad [\text{Ans: } -5496.4 \text{ KJ/mol}]$$

5. Glycerol ($\text{C}_3\text{H}_8\text{O}_3$) is a well known organic compound due to its versatile uses. Calculate the standard enthalpy of formation of Glycerol from the data given below.



$$[\text{Ans: } -669.6 \text{ KJ/mol}]$$

6. Draw a fully labeled Born Haber cycle for Rubidium chloride (RbCl) and determine the lattice energy by using the following values. (all in kJ/mol).

- I.P._{1st} of Rb = 403

- Electron affinity of Cl = -349 KJ/mol

- Bond energy of $\text{Cl}_2 = 242 \text{ KJ/mol}$

- Sublimation energy of Rb = 86.5 KJ/mol

- Heat of formation of $\text{RbCl}_{(s)} = -430.5 \text{ KJ/mol}$

$$[\text{Ans: } -692 \text{ KJ/mol}]$$